

Chapter 3 Homework

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Exercise Group 3.1

- 2) Prove that if l is any line, there is at least one point not on l .

Proof. If l is any line, there is at least one point not on l .

1. Let l be any line.
2. By postulate I-1 there exists at least 3 non-colinear points, call them A , B , and C .
3. Observe that if A , B , and C were all on l then they would be colinear.
4. Therefore, at least one of them does not lie on l . □

- 3) Both a and b can not represent lines because they intersect at two distinct points, and we have already shown that any two lines can not intersect at more than 1 point.

- 4) Prove that if l is any line and P is on l , then there is another point on l .

Proof. If l is any line and P is on l , then there is another point on l .

1. Let l be any line, and let P be a point on l .
2. By postulate I-3, l contains at least 2 points.
3. Therefore, there exists some other point on l aside from P . □

- 5) We are only guaranteed 3 non-colinear points in our plane so far, call them A , B , and C . Since we need two points to define a line, we are only guaranteed the lines through AB , AC , and BC for a total of 3 lines.

- 6) Intersect means to have a point in common.

- 7) Let l be a line. From I-1 we know that there exists a point not on l , and from I-3 we know that there exists at least 2 points on l .

- 8) We can conclude that either l or m is not a line since they intersect multiple times.

- 9) Since both l and m are lines, and since they have two points in common, they must be the same line. We are guaranteed this from I-2.

- 10) We can say that the two lines intersect, or that they are the same line.

- 11) Prove that there are at least two lines on each point.

Proof. If A is a point, then there are two lines containing it.

1. Let A be any point.
2. Observe that by I-1 we are guaranteed that there exists two other points, call them B and C , such that all three points are not colinear.
3. Observe that by I-2, there exists lines l and m through AB and AC respectively.
4. Therefore, there are 2 distinct lines containing A . □

- 12) We have no way of ensuring that $B \neq D$ and that $E \neq C$.

Exercise Group 3.2

- 1) $A_1 - A_2 - A_3 - A_4 - A_5$; the betweenness relations are $A_1A_2A_3, A_1A_2A_4, A_1A_2A_5, A_1A_3A_4, A_1A_3A_5, A_1A_4A_5, A_2A_3A_4, A_2A_3A_5, A_2A_4A_5, A_3A_4A_5$
- 2) If ABC and ACD , then the four points A, B, C , and D lie on one line.

Proof. If ABC and ACD , prove that the four points A, B, C , and D lie on one line.

1. Let points A, B, C , and D exist such that ABC and ACD .
 2. Since ABC , then by II-1 A, B , and C all lie on the same line, call it l .
 3. Similarly, since ACD then A, C , and D all lie on the same line, call it l' .
 4. From I-2, we know that there is a unique line containing the two points A and C .
 5. Therefore l and l' are the same lines, and so the four points A, B, C , and D lie on one line. □
- 4) We know CAD and ADB , so our picture is $-C - A - D - B-$. The other betweenness relations are CDB and CAB .
 - 5) We don't know any other betweenness relations. We could have $-C - A - D - B-$ or $-C - A - B - D-$.
 - 6) There are 4; $HEFG, EHFG, EFHG$, and $EFGH$.
 - 7) A and B are points on a line l . Write three true statements about points either on l or not on l . Give reasons.
 - 1) There is a point D on l such that ABD ; Postulate II-4
 - 2) There is a point C not on l ; Exercise 2 of Group 3.1
 - 3) There is a point E on l such that AEB ; Postulate II-4
 - 4) For any two points C and D not on l , they either lie on the same side or on opposite sides of l ; Postulate II-5
 - 8) This problem has 2 parts.
 - a) $V = A_1, R = A_2, S = A_3, T = A_4$
 - b) $B = A_1, C = A_2, A = A_3, D = A_4, E = A_5$
 - 9) Prove that there are at least 4 lines on a point.

Proof. There are at least 4 lines on a point.

1. Let P be any point, and observe that by I-1 there exists two other points A and B such that A, B , and P are non-collinear.
 2. Let l be the line through AB and observe that P does not lie on l .
 3. By II-4, there exists points C and D such that ACB and ABD .
 4. Then by II-3 we know that A, B, C , and D are all distinct points.
 5. Then the 4 lines through AP, BP, CP , and DP are distinct lines containing P (otherwise they'd intersect l twice).
 6. Therefore, there are at least 4 lines containing P . □
- 10) DEG and DGF
 - 11) II-1

- 12) Given that ABC and BCD , prove that ACD .

Proof. If ABC and BCD , then ACD .

1. Let ABC and BCD , and observe that II-3 guarantees us that there's a way to relabel the points A_1, A_2, A_3 , and A_4 such that $A_1A_2A_3$, $A_1A_2A_4$, $A_1A_3A_4$, and $A_2A_3A_4$ are our only betweenness relations.
2. Observe that either $ABC = A_2A_3A_4$ or $BCD = A_2A_3A_4$ since both ABC and BCD start and end with different points.
3. If we allow $ABC = A_2A_3A_4$, then $BCD = A_3A_4A_1$ which is not one of our betweenness relations.
4. Then it must be the case that $BCD = A_2A_3A_4$ and $A = A_1$.
5. Then from $A_1A_3A_4$ we get ACD . □

- 13) Given that ACB and CDB , prove that ACD and ADB . Then choose a point E such that DEB , and list all the betweenness relations.

Proof. If ACB and CDB , then ACD and ADB .

1. Let ACB and CDB , and observe that II-3 guarantees us that there's a way to relabel the points A_1, A_2, A_3 , and A_4 such that $A_1A_2A_3$, $A_1A_2A_4$, $A_1A_3A_4$, and $A_2A_3A_4$ are our only betweenness relations.
2. Observe that either $ACB = A_2A_3A_4$ or $CDB = A_2A_3A_4$ since both ACB and CDB start with different points and end with the same point.
3. If we allow $ACB = A_2A_3A_4$, then $CDB = A_3A_1A_4$ which is not one of our betweenness relations.
4. Then it must be the case that $CDB = A_2A_3A_4$ and $A = A_1$.
5. Then $A_1A_2A_3 = ACD$ and $A_1A_3A_4 = ADB$. □

If we add E such that DEB , then our new picture is $A - C - D - E - B$, and we get the betweenness relations ACE , ADE , CDE , AEB , CEB , and DEB . We can carry out this process indefinitely demonstrating that there are infinitely many points in a line segment.

- 14) Let l be a line, let A be a point on one side and B be on the other side of the line, and let C be on the opposite side of the line as A . Prove that B and C are on the same side of the line.

Proof. If l is a line, A is a point on one side of the line, B is on the other side of the line, and C is on the opposite side of the line as A , then B and C are on the same side of the line.

1. Let l be a line, let A be a point on one side and B be on the other side of the line, and let C be on the opposite side of the line as A .
2. From II-5, we know that l divides the points not on l into two sides called the sides of l .
3. Let X be the side such that $A \in X$ and let Y be the other side such that $B \in Y$.
4. Since C is on a side of l then we must have $C \in X \vee C \in Y$.
5. Since C is not on the same side as A , we know $C \notin X$.
6. Therefore, $C \in Y$, and so C is on the same side as B ; in other words, both C and B are in Y . □

- 15) A line l does not contain any vertices of triangle $\triangle ABC$. The line intersects the side AB , but does not intersect the side BC . Prove that l must intersect AC .

Proof. If a line l does not contain any vertices of triangle $\triangle ABC$, intersects the side AB , and does not intersect the side BC , then it intersects AC .

1. Let l be a line that does not contain any vertices of triangle $\triangle ABC$, intersects AB , and does not intersect BC .
2. Observe that l splits the plane into two sides, and that the three points A, B , and C do not lie on l .
3. Since l intersects AB , then there must be a point D such that D is on l and ADB .
4. Then A and B must lie on opposite sides of l .
5. Observe that if B and C were to lie on opposite sides of l , then l would have to intersect the segment BC .
6. Therefore, both B and C are on the same side, and are on the opposite side of A .
7. Since A and C lie on opposite sides of l , there must exist a point E on l such that AEC .
8. Therefore, l intersects AC . □

- 16) Let A and B be two points on the same side of line l . Choose C such that ACB . Prove that C and A are on the same side of l .

Proof. If A and B are on the same side of l , and C is a point such that ACB , then A and C are on the same side of l .

1. Let A and B be points on the same side of l , and let C such that ACB .
2. Assume BWOC (by way of contradiction) that A and C are on opposite sides of l .
3. Then there exists D on l such that ADC .
4. Then from ACB and ADC we get ADB , showing that A and B are on opposite sides of $l \Rightarrow \Leftarrow$ (contradiction).
5. Therefore, A and C are on the same side of l . □

- 17) If A is on line l and B is not on l and ACB , prove that C and B are on the same side of l .

Proof. If A is on line l and B is not on l and ACB , the C and B are on the same side of l .

1. Let A be a point on line l , B be a point not on l , and C be a point such that ACB .
 2. Assume BWOC that C and B are on opposite sides of l .
 3. Then there exists a point D on l such that CDB .
 4. Then from ACB and CDB we get ADB , and so the points A , D , and B lie on a single line.
 5. Then the line through A , D , and B intersects l at two points, namely A and $D \Rightarrow \Leftarrow$.
- Therefore C and B are on the same side of l . □

- 18) Prove that for every line l there is a point on each side of l .

Proof. If l is a line, then there is a point on either side of l .

1. Let l be a line and let O be a point on l .
2. Observe that by I-1 there must be a point not on l , call it A .
3. Then by I-2 there's a line l' through the points A and O .
4. Then by II-4 there must exist a point B on l' such that AOB .
5. Therefore A and B are points on opposite sides of l . □

Exercise Group 3.3

- 1) We mean the set of all points Q such that if $A \in Q$ and $R \in r$ where $R \neq P$, then the segment QR does not intersect l .

Exercise Group 3.4

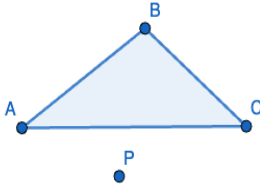
- 2) Infinite.
- 3) A straight angle.
- 4) They are supplementary angles.
- 5) 6 with 2 pairs of vertical angles.
- 6) 15 with 6 pairs of vertical angles.
- 7) $>$ $<$
- 8) Draw one angle interior to another angle, with one of their sides being common.
- 10) There is only one, and it is the angle that emerges by replacing the rays with lines.
- 11) Because being a vertical angle is a relation between two angles, and is not an intrinsic property of an angle.

Exercise Group 3.5

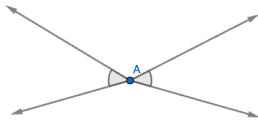
- 2) .
 - a) $\angle AOB$
 - b) $\angle BOC$
 - c) $\angle AOD$

Exercise Group 3.6

- 1) Yes.
- 2) The point P can be as illustrated.



- 3) The two angles' interiors have no points in common.



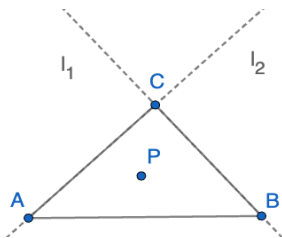
- 4) Because one of the rays isn't on a side of the line containing the other ray.
- 6) No, P could lie on segment BC of the triangle.
- 7)
 - a) the left side of m
 - b) the bottom side of l
 - c) the right portion of l
 - d) the top portion of m
 - e) the top side of l
 - f) the interior of the angle

- 8) Prove that the interior of $\triangle ABC$ is equivalent to the interior of $\angle A$ intersected with the interior of $\angle B$.

Proof. A point is in the interior of $\triangle ABC$ iff it lies in the interior of $\angle A$ intersected with the interior of $\angle B$.

1. ($\Rightarrow$) Let P be any point in the interior of $\triangle ABC$.
2. Then it is in the interior of all three angles, including $\angle A$ and $\angle B$.
3. ($\Leftarrow$) Let P be any point in the interior of $\angle A$ and the interior of $\angle B$.
4. Let l_2 be the line through AC and l_1 be the line through BC .
5. Since P is in the interior of $\angle A$, then P is on the same side of l_2 as B .
6. Since P is in the interior of $\angle B$, then P is on the same side of l_1 as A .
7. Then P is in the interior of $\angle C$ as well.
- 8 Therefore P is in the interior of $\triangle ABC$.

□



- 9) In $\angle AOB$, not a straight angle, point P is between A and B . Prove that P is in the interior of the angle.

Proof. If $\angle AOB$ is not straight and APB , then P is in the interior of $\angle AOB$.

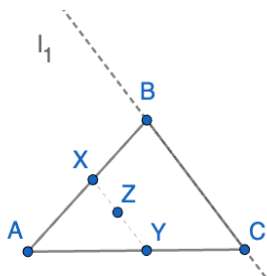
1. Let $\angle AOB$ be a non-straight angle, and let P be a point such that APB .
2. Then P is on the same side of the line through OA as B , and the same side of line through OB as A .
3. Therefore P is in the interior of $\angle AOB$.

□

- 10) In triangle ABC we have the relations AXB , AYC , and XZY . Prove that Z is in the interior of the triangle.

Proof. If AXB and AYC on $\triangle ABC$, then Z is in the interior of $\triangle ABC$.

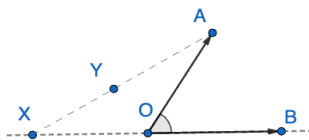
1. Let AXB and AYC on $\triangle ABC$, and let l_1 be the line through BC .
2. Since X lies on ray AB and Y lies on ray AC , then $\angle XAY = \angle A$.
3. Since XZY , then Z is in the interior of $\angle XAY = \angle A$, and it remains to show that Z is in the interior of $\angle B$.
4. Since Z is in the interior of $\angle A$, then it is on the same side of the line through AB as C .
5. It remains to show that Z is on the same side of l_1 as A .
6. Assume BWOC that Z is on the opposite side of l_1 as X and Y .
7. Then there exists points R_1 and R_2 on l_1 such that XR_1Z and ZR_2Y .
8. From XR_1Z , ZR_2Y , and XZY , we know that all 5 points X , R_1 , Z , R_2 , and Y lie on a single line, and this line intersects l_1 twice (at R_1 and R_2) $\Rightarrow \Leftarrow$.
9. Therefore Z lies on the same side of l_1 as X and Y , and so lies on the same side of l_1 as A .
10. Therefore Z lies in the interior of $\angle B$, and so lies in the interior of $\triangle ABC$. □



- 11) If $\angle AOB$ is not a straight angle and we have the relations XOB and XYA , prove that Y is in the exterior of $\angle AOB$.

Proof. If $\angle AOB$ is not a straight angle, XOB , and XYA , then Y is in the exterior of $\angle AOB$.

1. Let $\angle AOB$ be a non-straight angle, let XOB , and let XYA .
2. Since XOB , then X is on the opposite side of the line through AO as B .
3. Since XYA , then Y is on the same side of the line through OA as X , and so is on the opposite side of the line through OA as B .
4. Therefore Y so is not in the interior of $\angle AOB$ and does not lie on ray OB .
5. Observe that if Y lied on ray OA , then the line through XYA would intersect the line through OA twice.
6. Therefore Y lies in the exterior of $\angle AOB$. □



- 12) I proved this in Theorem 3-6
 13) I proved this in Theorem 3-6
 14) I proved this in Theorem 3-6

Chapter 3 Algebra Review

- 1) If $c = p - q$, then by definition of subtraction $p = q + c$.
- 2) $7 - 3 = 4$ because $7 = 3 + 4$. If $7 - 3 = x$, then $7 = x + 3$. So x is the unique number s.t. when it's added with 3, it sums to 7 (namely 4).
- 3) $\frac{9}{5} - \frac{3}{5} = \frac{6}{5}$, since $\frac{3}{5} + \frac{6}{5} = \frac{9}{5}$. $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, since $\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$. $\frac{7}{8} - \frac{3}{4} = \frac{1}{8}$, since $\frac{3}{4} + \frac{1}{8} = \frac{7}{8}$.
 $4.878 - 2.589 = 2.289$ since $2.589 + 2.289 = 4.878$.
- 4) $-7 = 0 - 7$ by definition.

- 5) Let $a = 9 - 3 \Rightarrow 9 = 3 + a$, then $9 + (-3) = 3 + a + (-3) = a + 3 + (-3) = a + 0 = a$, and so $9 - 3 = a = 9 + (-3)$.
 Let $a = 11 - 8 \Rightarrow 11 = 8 + a$, then $11 + (-8) = 8 + a + (-8) = a + 8 + (-8) = a + 0 = a$, and so $11 - 8 = a = 11 + (-8)$.
 Let $a = \frac{5}{6} - \frac{1}{3} \Rightarrow \frac{5}{6} = \frac{1}{3} + a$, then $\frac{5}{6} + (-\frac{1}{3}) = \frac{1}{3} + a + (-\frac{1}{3}) = a + \frac{1}{3} + (-\frac{1}{3}) = a + 0 = a$, and so $\frac{5}{6} - \frac{1}{3} = a = \frac{5}{6} + (-\frac{1}{3})$.

- 6) Prove that $a + (-b) = a - b$.

Proof. $a + (-b) = a - b$.

1. Let $x = a - b$, and observe that $a = b + x$.

2. Then $a + (-b) = (b + x) + (-b) = (x + b) + (-b) = x + [b + (-b)] = x + 0 = x$.

3. Then $a - b = x = a + (-b)$. □

- 7) Let $a = -(7+3) \Rightarrow a = 0 - (7+3) \Rightarrow 0 = a + (7+3)$. Then $(-7) + (-3) = (-7) + (-3) + 0 = (-7) + (-3) + a + (7+3) = a + [(-7) + 7] + [(-3) + 3] = a + 0 + 0 = a$. Therefore $-(7+3) = a = (-7) + (-3)$.

Let $a = -(2+7) \Rightarrow a = 0 - (2+7) \Rightarrow 0 = a + (2+7)$. Then $(-2) + (-7) = (-2) + (-7) + 0 = (-2) + (-7) + a + (2+7) = a + [(-2) + 2] + [(-7) + 7] = a + 0 + 0 = a$. Therefore $-(2+7) = a = (-2) + (-7)$.

- 8 Show that $-(a+b) = (-a) + (-b)$.

Proof. $-(a+b) = (-a) + (-b)$

1. Let $x = -(a+b) \Rightarrow x = 0 - (a+b) \Rightarrow x + (a+b) = 0$ or $0 = x + (a+b)$.

2. Then $(-a) + (-b) = (-a) + (-b) + 0 = (-a) + (-b) + [x + (a+b)] = x + [(-a) + a] + [(-b) + b] = x + 0 + 0 = x$.

3. Then $-(a+b) = x = (-a) + (-b)$. □

- 9 Assume $x = a/0$ is defined. Then $a = 0 \cdot x$. Since $0 \cdot x = 0$, then $a = 0$. Observe then that for any number x , $x = 0/0$ since $0 = 0 \cdot x$. Since $0/0$ isn't any particular value, then it is undefined.

- 10 $\frac{3}{4}$ is the number such that $1 = (\frac{4}{3} \cdot \frac{3}{4})$
 $-\frac{2}{3}$ is the number such that $1 = (-\frac{3}{2} \cdot -\frac{2}{3})$
 $\frac{4}{-5}$ is the number such that $1 = (\frac{-5}{4} \cdot \frac{4}{-5})$
 $\frac{7}{16}$ is the number such that $1 = (\frac{16}{7} \cdot \frac{7}{16})$
 $\frac{5}{10}$ is the number such that $1 = (\frac{10}{5} \cdot \frac{5}{10})$
 $\frac{1}{1}$ is the number such that $1 = (\frac{1}{1} \cdot \frac{1}{1})$
 $\frac{2}{2}$ is the number such that $1 = (\frac{2}{2} \cdot \frac{2}{2})$

- 11) $\frac{a}{b} = \frac{a}{b}$, and then by definition of division, $a = b \cdot \frac{a}{b}$.

- 12) Prove that $a/b + c/b = (a+c)/b$.

Proof. $\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}$, where $b \neq 0$.

1. Let $x = \frac{a}{b}$, $y = \frac{c}{b}$, and $z = \frac{a+c}{b}$, where $b \neq 0$.

2. Then $a = x \cdot b$, $c = y \cdot b$, and $a + c = z \cdot b$.

3. Then $z \cdot b = a + c = (x \cdot b) + (y \cdot b) = (x + y) \cdot b$.

4. Since $b \neq 0$, then $z = x + y$.

5. Substituting back in we get $\frac{a+c}{b} = \frac{a}{b} + \frac{c}{b}$. □

- 13) Let $x = \frac{1}{2}$. Then $2x = 1 \rightarrow (2)(2x) = 2(1) \rightarrow 4x = 2 \rightarrow x = \frac{2}{4}$.

- 14) Prove that if $b \neq 0$ and $c \neq 0$, then $a/b = ac/bc$.

Proof. If $b \neq 0$ and $c \neq 0$, then $\frac{a}{b} = \frac{ac}{bc}$.

1. Assume $b \neq 0$, $c \neq 0$, and let $x = \frac{a}{b}$.

2. Then $a \cdot x = b$, and so $(ax) \cdot c = bc \iff (ac) \cdot x = bc$.

3. Since $b \neq 0$ and $c \neq 0$, then $bc \neq 0$.

4. Then from $(ac) \cdot x = bc$ we get that $x = \frac{bc}{ac}$.

5. Therefore $\frac{a}{b} = x = \frac{bc}{ac}$. □